Lipid travels two routes: motors at a fixed speed both ways, and the stream

Nile Red lipid moves at a steady 2.7 μm/s in both directions, whatever the hypha's flow does: these are motors. Motors carry about 43 % of the moving lipid (30 plates, 2,233 edges), mostly against the stream and mostly tip-ward; the rest rides the pressure-driven stream.

☐ Decide placement: one main-text paragraph (Fig 2 or Discussion) plus this SI section, or SI only. Write the main-text paragraph.

Data and gates

Five paper plates (bright-field 20 fps, Nile Red 2 fps); the 10 and 20 fps fluorescence tests; 30 extension plates (LIPIDS_2024 and heat-shock sessions) run through `fl_extra_driver.sh`. Videos with the lid off are screened out by the bright-field swing (≤ 1.5 μm/s, `kymo_stability.py`); fluorescence edges enter only with visible moving signal, r(0.5 s) ≥ 0.1 (`fl_signal_gate.py`): 2,285 of 5,476 FL edges.

☐ Table of plates: session, condition (`plate_conditions.tsv`), videos, gated edges. Temperature unknown for 9 heat-shock plates; CHC2408A287 kept as a control unconfirmed.

☐ Methods paragraph: streak detection settings (FL std 0.20/0.35 with imax normalisation, BF 0.10/0.17, ≥ 0.75 s).

Steady streaks at motor speed do not follow the flow

Fluorescent streaks cluster near 2.6 μm/s in both directions, and their speed does not change with the edge's stream speed (slope ≈ 0). On synthetic kymographs the fluorescence detector reads speeds without bias; bright-field reads slow.

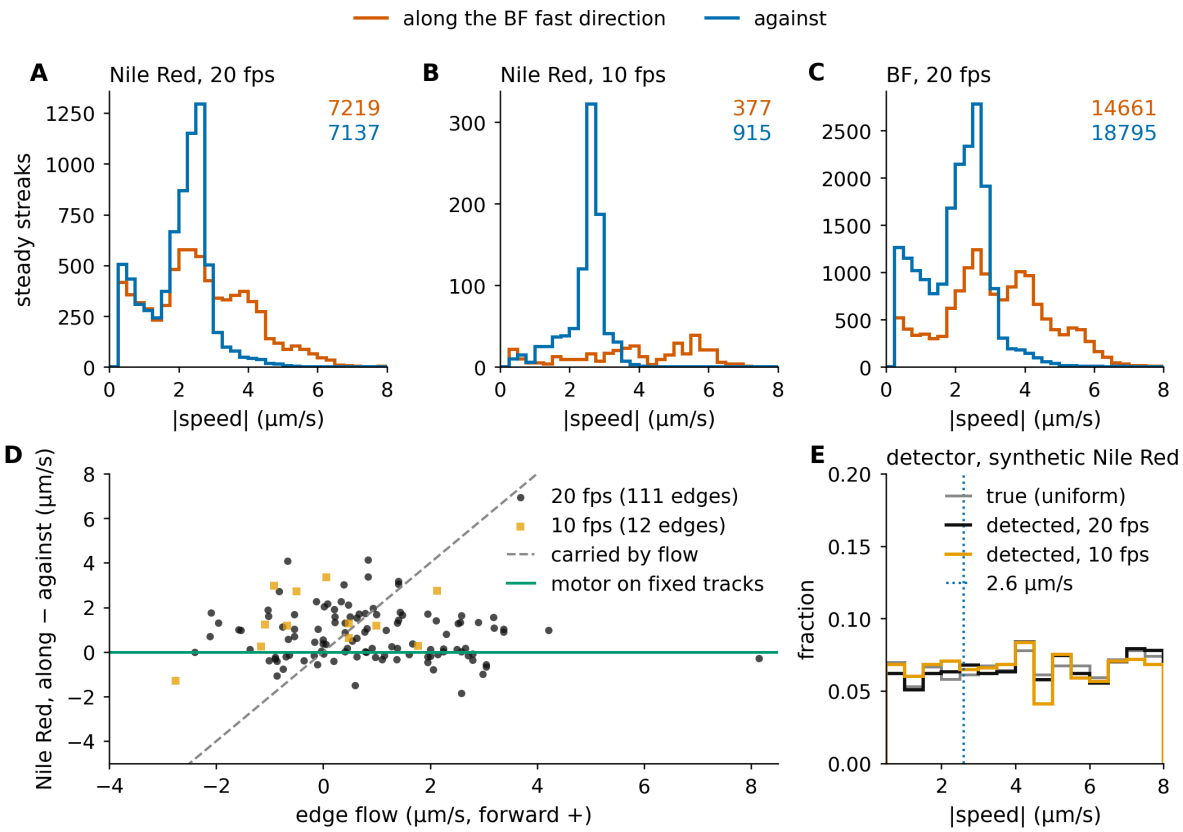


Figure S1: Steady fluorescent streaks run at motor speed in both directions, independent of the edge's flow. (src: `manuscript_figures/figS_motor_speed.py`)

☐ Caption with panel letters and n; state the synthetic check in one sentence.

Motors carry about 43 % of the moving lipid

A mixture fit per edge (motors at $\pm v_m$, a stream at the bright-field fast speed, slow drift, noise), weighted by streak brightness, splits the lipid between motors and stream. Per edge the motor share spans 0 to 1; pooled it is 0.43, per plate 0.22–0.75. Most motor lipid runs against the stream (per plate 0.5–0.9).

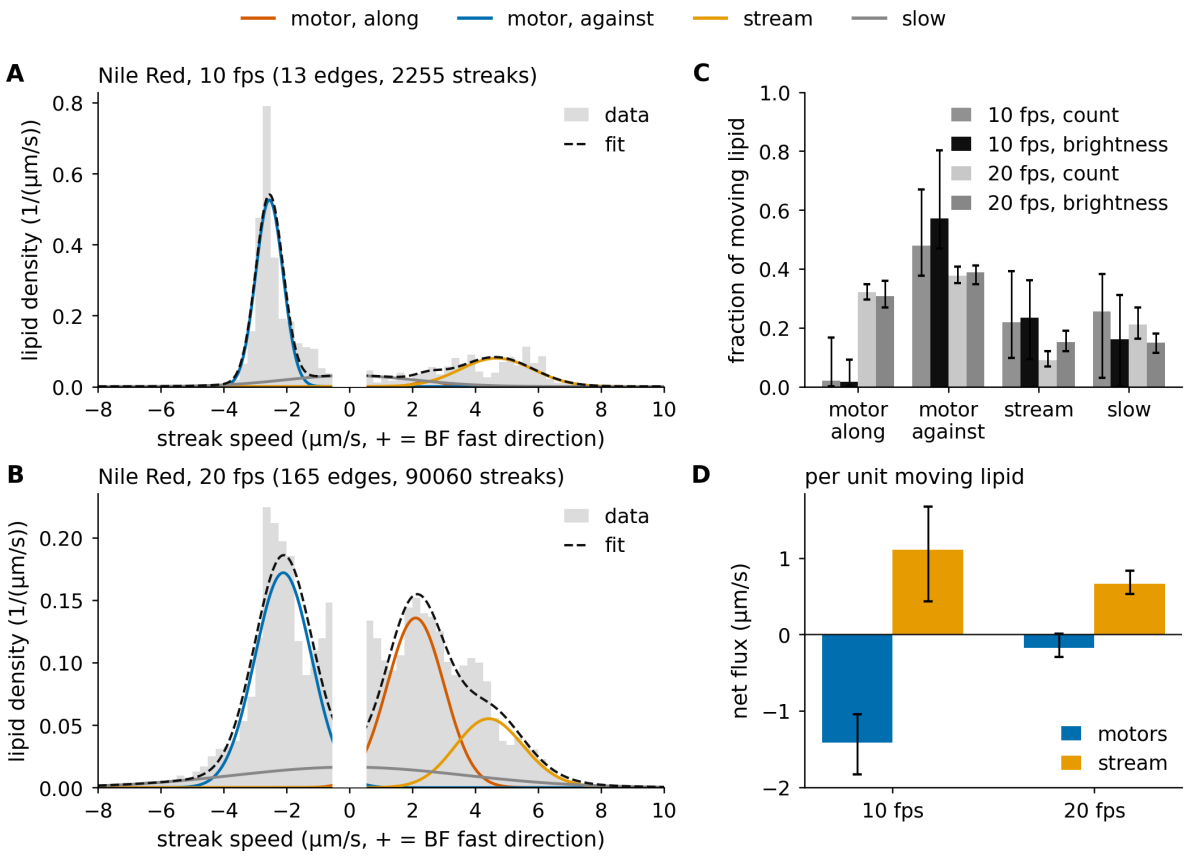


Figure S2: Mixture fit: motor and stream components of the fluorescent streak speeds. (src: `manuscript_figures/figS_motor_mixture.py`)

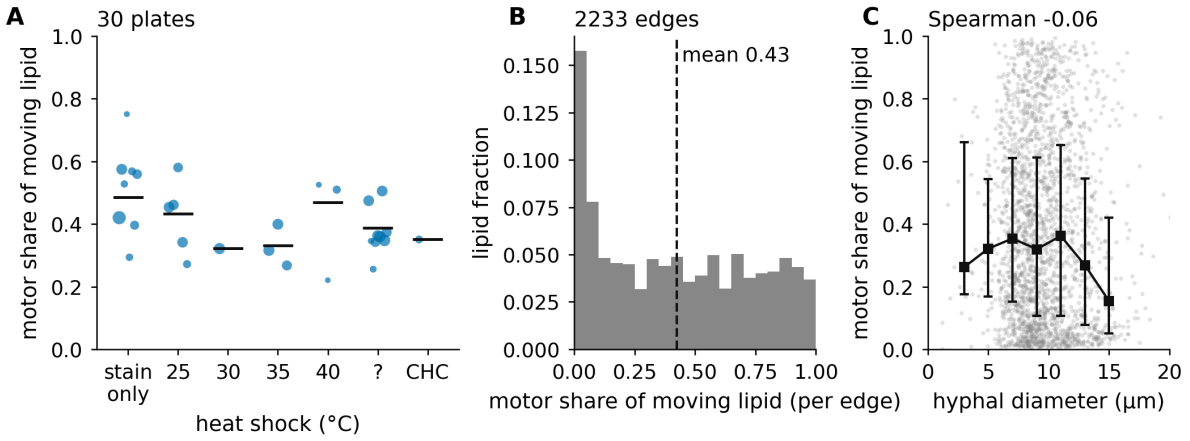


Figure S3: Motor share of the moving lipid on the 30 extension plates: per plate by condition, per edge, and against hyphal diameter. (src: `manuscript_figures/figS_motor_share.py`)

☐ Normalisation: is brightness proportional to lipid amount? State the assumption or test it (bleaching, depth).

The share does not follow diameter or temperature

Motor share against hyphal diameter (lumen from the bright-field U-Net): Spearman −0.06 (2,233 edges); flat at 0.32–0.36 from 4 to 12 µm, lower only above 14 µm (0.15, 92 edges). By heat-shock temperature the groups (0.32–0.47) differ less than plates within a group.

☐ Re-test plate age on all 30 plates (ρ +0.32 on the first set only).

☐ Edge type (runner, offshoot, …) and distance to the tip need network stores; none exist for the 30 extension plates.

Motors run mostly tip-ward; the root-ward material at motor speed is not mirrored motors
Oriented by the growth graph on the five paper plates, 0.65 of the bright-field motor material and 0.60 of the Nile Red motor material moves tip-ward; the stream runs mostly root-ward (0.38 tip-ward). A mixture fit on the oriented speeds puts the root-ward material near motor speed in its own broad population (2.80 ± 0.97 µm/s), not in motors tied to the tip-ward ones (0.002 of the streaks; symmetric motors worse by ΔBIC 3,249). Growth labels that point at the roots (8 % of edges) carry the same signature; flipping them moves the motor share by 0.03 (`orientation_flag.py`).

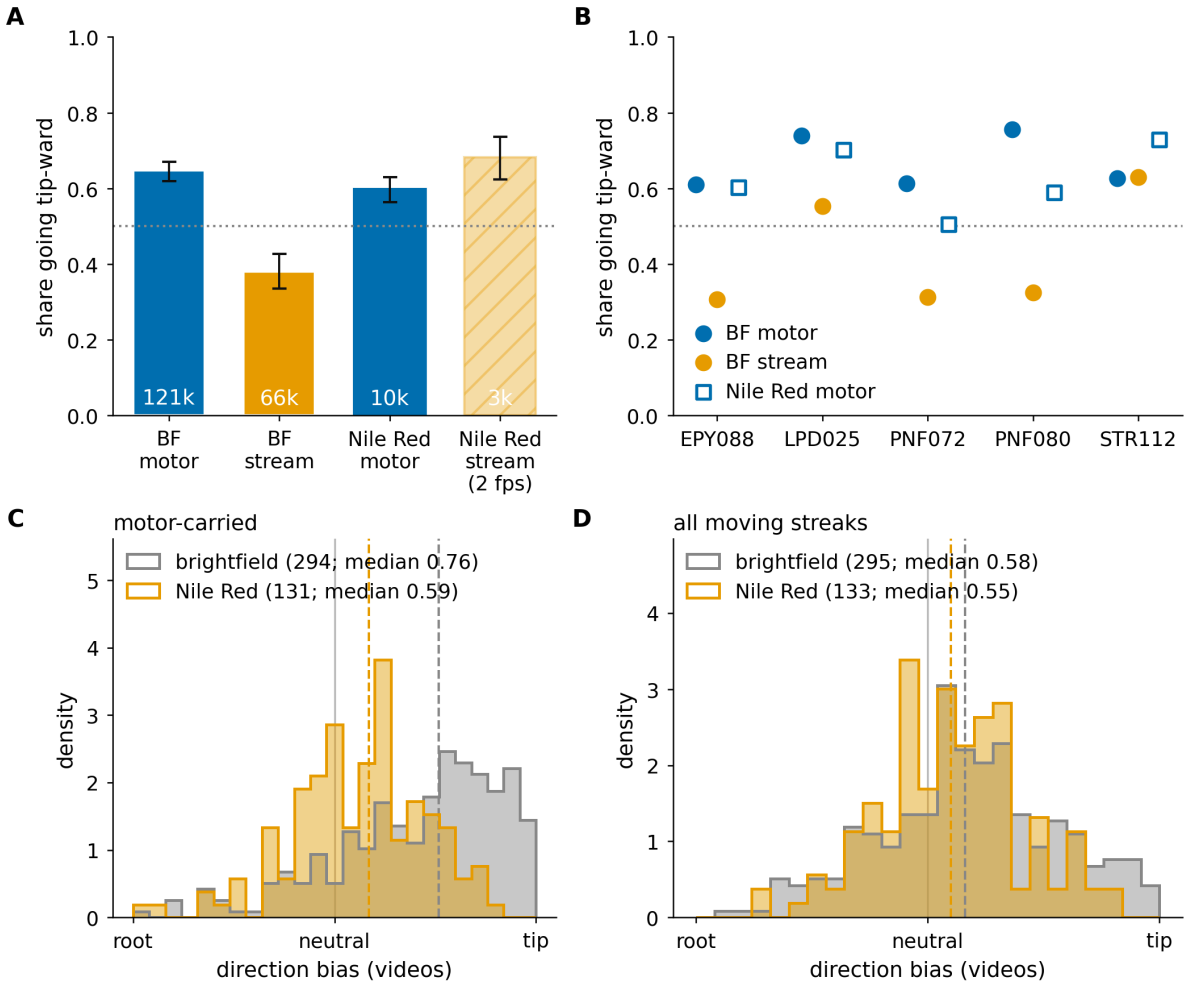


Figure S4: Tip-ward share of motor and stream material, and per-video direction bias. (src: `manuscript_figures/figS_lipid_direction_oriented.py`)

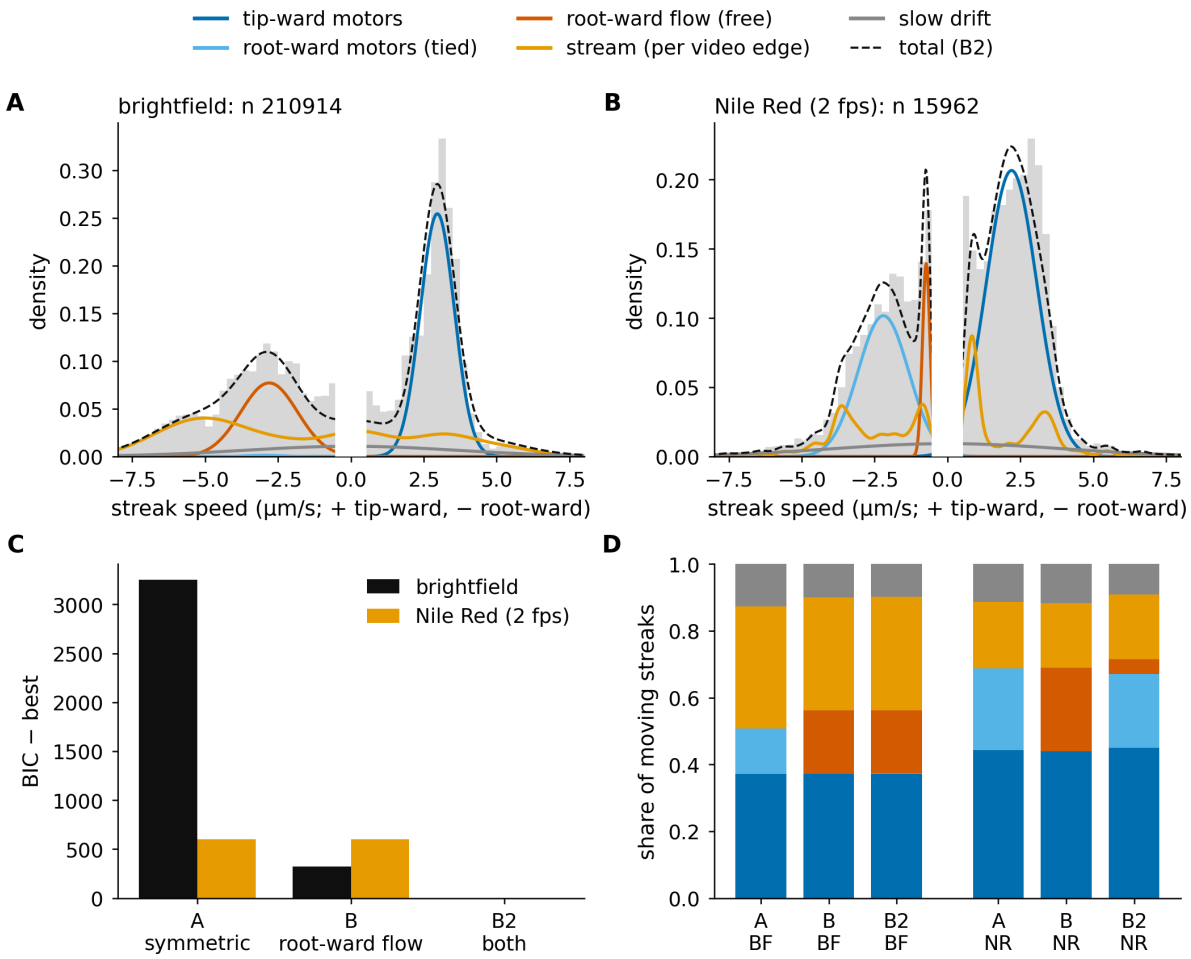


Figure S5: Oriented mixture: the root-ward material near motor speed is its own population. (src: `manuscript_figures/figS_oriented_mixture.py`)

The two counts in this supplement measure different things. "Motor direction" counts edges where exactly one stream is motor-like and asks which way it runs (against growth on 13 %); here every motor-speed streak counts. On the 15 model stores (1,794 video edges with ≥ 10 motor-speed streaks) the root-ward share is 0.35 per streak, but the motor majority runs root-ward on only 0.24 of edges, and on 0.20 of the 530 edges with exactly one motor-like direction. Most root-ward motor-speed material (64 %) sits on edges where neither direction is motor-dominated: a minority at motor speed inside a stream, which an edge-level rule cannot see and which matches the broad root-ward population of the oriented fit.

☐ The 13 % (765 edges, `motor_disagreement.py`) against 20 % here (530 edges, motor share > 0.5 per direction): align the motor-like rule, or name both.

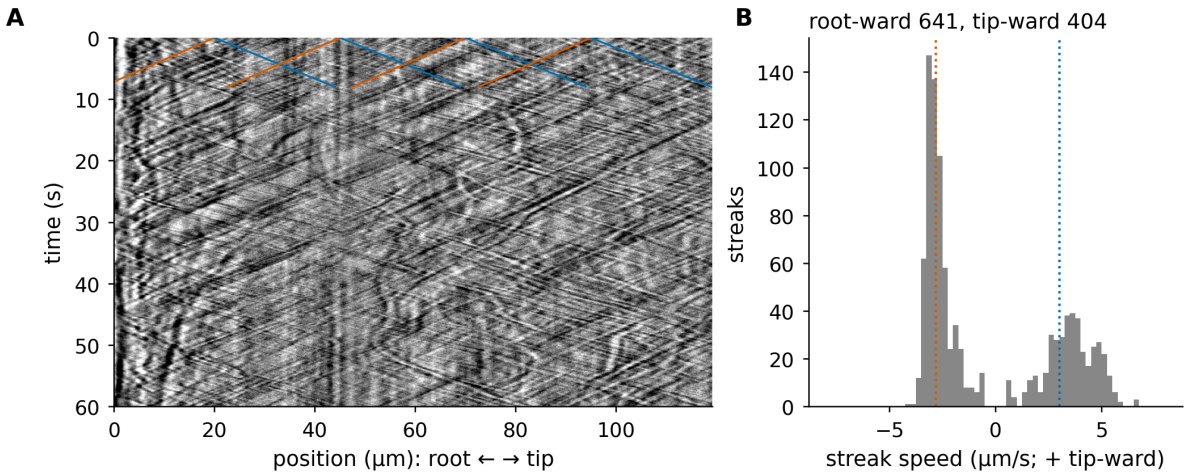


Figure S6: One edge with root-ward material at motor speed (PNF2307A072, bright-field 10 fps): sharp root-ward streaks at 2.8 µm/s, a broad tip-ward population. (src: `manuscript_figures/figS_rootward_example.py`)

☐ What is the root-ward population physically (fast return flow in thin hyphae, retrograde motors with a broad load)? In the example it is the sharp peak and the tip-ward side is broad: say which side looks like motors.

☐ Nile Red at 2 fps cannot separate these models (ΔBIC ≈ 600 for both alternatives): say so.

The stream left after removing motors matches the model's backflow

On the 15 stores of the main model, the decomposed stream speed on edges where its direction matches the model's return flow (60 %) follows |bflow_flux|/(πa²): ρ +0.39, median ratio 1.05. Removing motors lets the measured pressure speed reach lower values than the per-direction median.

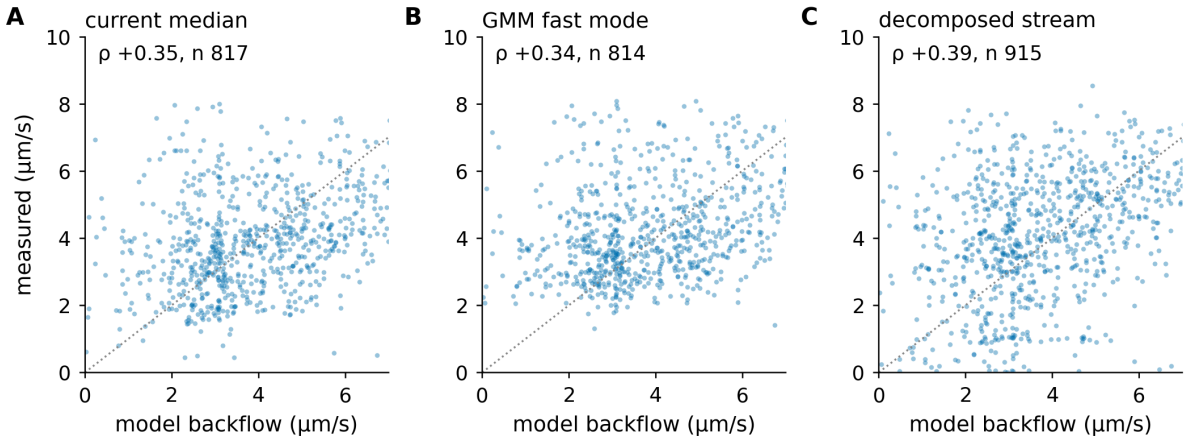


Figure S7: Measured backflow speed against the model's on direction-matched edges: the per-direction median, the GMM fast mode and the decomposed stream. (src: `manuscript_figures/figS_stream_model.py`)

☐ Say why 40 % of edges mismatch: they carry little model demand (F median 88 against 6,200 on matched edges).

Scoring the main model's to-root speed with the decomposed stream instead of the median changes little: on the same 915 matched edges ρ +0.39 against +0.38, median ratio 1.05 against 0.95, within 2× 0.76 against 0.82 (the decomposed stream reaches lower speeds); filling the unmatched edges with

the median gives ρ +0.35 against +0.34 on all 1,524. The model's accuracy does not depend on the read-out; the main figures keep the median.

Limit: fast lipid in the stream is not resolved

At 20 fps the share of fast fluorescent streaks stays at the noise floor (0.045 against 0.040 in the control), so the lipid riding the fast root-ward stream is under-counted and the motor share is an upper bound.

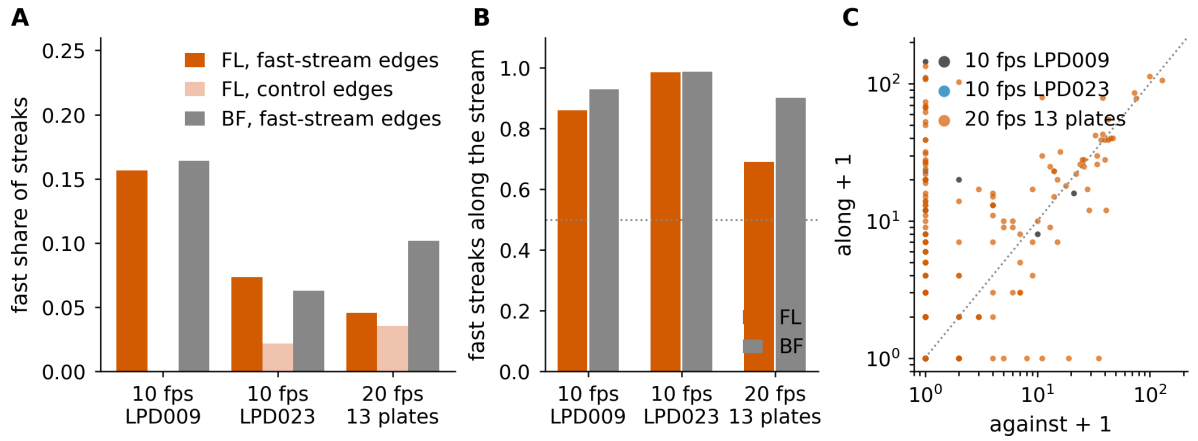


Figure S8: Fast fluorescent streaks (≥ 5 µm/s) at 10 and 20 fps against the noise control. (src: `manuscript_figures/figS_fl_fast.py`)

☐ Reconcile with 10 fps: on LPD009 fast FL streaks stand well above the control (0.156 against 0.000) and run with the stream (0.86); on LPD023 0.073 against 0.022. So the limit is the 20 fps set (lid, focus, plates), not the frame rate alone. Reword the claim.

☐ Lid-on 20 fps fluorescence videos with stable streams would settle the fast lipid: list candidates.

References